

VPC 2

1) Create one VPC, with 1 public subnet and 1 private subnet.

Create VPC

- Login → AWS Console
- Search → **VPC**
- Click → **Create VPC**
- Select → **VPC only**
- Enter:
- Name: MyVPC
- IPv4 CIDR: 198.162.0.0/16
- Click → **Create VPC**

Name: MyVPC

IPv4 CIDR block: 192.168.0.0/16

Now your VPC IP range will be:

- 192.168.0.0 → 192.168.255.

The screenshot displays the AWS Management Console interface. At the top, the navigation bar shows the AWS logo, a search bar, and the region 'Europe (Stockholm)'. The breadcrumb trail indicates the path: VPC > Your VPCs > Create VPC.

The main content area is titled 'Create VPC' with an 'Info' link. Below the title, a description states: 'A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.'

The 'VPC settings' section contains the following configuration options:

- Resources to create:** 'VPC only' is selected, with 'VPC and more' as an alternative.
- Name tag - optional:** The value 'my-vpc-01' is entered.
- IPv4 CIDR block:** 'IPv4 CIDR manual input' is selected. The 'IPv4 CIDR' field contains '198.162.0.0/16'. A note specifies: 'CIDR block size must be between /16 and /28.'
- IPv6 CIDR block:** 'No IPv6 CIDR block' is selected. Other options include 'IPAM-allocated IPv6 CIDR block', 'Amazon-provided IPv6 CIDR block', and 'IPv6 CIDR owned by me'.

A green success message at the top of the details section reads: 'You successfully created vpc-01438603bad60b405 / my-vpc-01'.

The details section for 'vpc-01438603bad60b405 / my-vpc-01' includes an 'Actions' dropdown menu and a 'Details' tab. The details are organized into four columns:

Details	State	Block Public Access	DNS hostnames
VPC ID vpc-01438603bad60b405	Available	Off	Disabled
DNS resolution Enabled	Tenancy default	DHCP option set dopt-0d1145d6488ab4338	Main route table rtb-09855a9ec4d3e985f
Main network ACL acl-0bae2b69e49696cb9	Default VPC No	IPv4 CIDR 198.162.0.0/16	IPv6 pool -
IPv6 CIDR (Network border group) -	Network Address Usage metrics Disabled	Route 53 Resolver DNS Firewall rule groups -	Owner ID 299582146630
Encryption control ID -	Encryption control mode -		

At the bottom, a horizontal menu provides links to 'Resource map', 'CIDRs', 'Flow logs', 'Tags', 'Related resources', and 'Integrations'.

STEP 2: Create Public Subnet

1. Go → Subnets → Create subnet

2. Select → MyVPC

3. Enter:

Name: Public-Subnet-01

CIDR block: 192.168.1.0/24

Click Create

This subnet range:

192.168.1.0 → 192.168.1.255

The screenshot shows the AWS Management Console interface. The top navigation bar includes the AWS logo, a search bar, and the user's profile (Srihari). The main content area is titled 'Create subnet' and shows the 'VPC' section with the VPC ID 'vpc-01438603bad60b405 (my-vpc-01)'. The 'Subnet settings' section shows the 'Subnet name' as 'public-subnet-01'. A green notification banner at the top of the 'Subnets' list states: 'You have successfully created 1 subnet: subnet-015068d839a347954'. The 'Subnets (1)' table lists the following subnet:

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
public-subnet-01	subnet-015068d839a347954	Available	vpc-01438603bad60b405 my-...	Off	192.168.1.0/24

STEP 3: Create Private Subnet

1. Click → Create subnet

2. Enter:

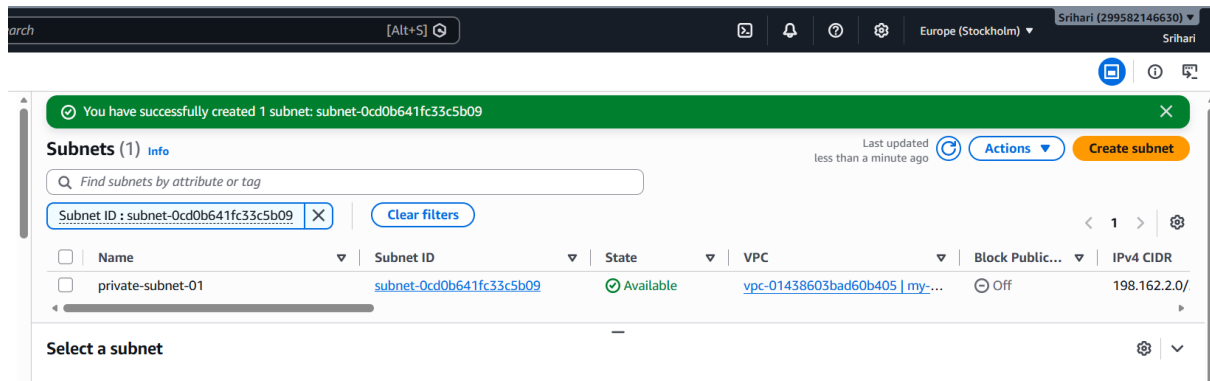
Name: Private-Subnet-01

CIDR block: 192.168.2.0/24

Click Create

This subnet range:

192.168.2.0 → 192.168.2.255



FINAL STRUCTURE

VPC: 192.168.0.0/16

Public Subnet: 192.168.1.0/24

Private Subnet: 192.168.2.0/24

STEP 4: Create Internet Gateway

1. Go → Internet Gateway
2. Click → Create
3. Name: MyIGW
4. Click → Create
5. Click → Attach to VPC → select MyVPC

[Alt+S]
Europe (Stockholm)
Srihari (299582146630)
Srihari

igw-08947989aa30b850c

The following internet gateway was created: igw-08947989aa30b850c - my-internet-gateway. You can now attach to a VPC to enable the VPC to communicate with the internet.
Attach to a VPC

igw-08947989aa30b850c / my-internet-gateway
Actions

Details
Info

Internet gateway ID
igw-08947989aa30b850c

State
Detached

VPC ID
-

Owner
299582146630

Tags (1)
Manage tags

Search tags

Key | Value

Name | my-internet-gateway

aws
Search
[Alt+S]
Europe (Stockholm)
Srihari (299582146630)
Srihari

VPC > Internet gateways > Attach to VPC (igw-08947989aa30b850c)

Attach to VPC (igw-08947989aa30b850c)
Info

VPC

Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs

Attach the internet gateway to this VPC.

Search vpc-01438603bad60b405

AWS Command Line Interface command

Cancel
Attach internet gateway

[Alt+S]
Europe (Stockholm)
Srihari (299582146630)
Srihari

igw-08947989aa30b850c

Internet gateway igw-08947989aa30b850c successfully attached to vpc-01438603bad60b405

igw-08947989aa30b850c / my-internet-gateway
Actions

Details
Info

Internet gateway ID
igw-08947989aa30b850c

State
Attached

VPC ID
vpc-01438603bad60b405 | my-vpc-01

Owner
299582146630

Tags (1)
Manage tags

Search tags

Key | Value

Name | my-internet-gateway

5. Create Public Route Table

1. Go → Route Tables
2. Click → Create
3. Name: PublicRT
4. Select VPC: MyVPC

Click Create

rch [Alt+S] Europe (Stockholm) Srihari (299582146630) Srihari

rtb-024d66a4ac6c45616

Route table rtb-024d66a4ac6c45616 | publicRT was created successfully.

rtb-024d66a4ac6c45616 / publicRT [Actions](#)

Details [Info](#)

Route table ID rtb-024d66a4ac6c45616	Main No	Explicit subnet associations -	Edge associations -
VPC vpc-01438603bad60b405 my-vpc-01	Owner ID 299582146630		

[Routes](#) [Subnet associations](#) [Edge associations](#) [Route propagation](#) [Tags](#) [Related resources](#)

Routes (1) [Both](#) [Edit routes](#)

Destination	Target	Status	Propagated	Route Origin
198.162.0.0/16	local	Active	No	Create Route Table

6. Add Internet Route

1. Select PublicRT
2. Click → Routes → Edit routes
3. Add:

Destination:

0.0.0.0/0

Target:

Internet Gateway → MyIGW

Save

[Alt+S]

Europe (Stockholm)
Srihari (299582146630)

VPC > Route tables > rtb-024d66a4ac6c45616 > Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin
198.162.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway	-	No	CreateRoute

Add route

Cancel Preview Save changes

Updated routes for rtb-024d66a4ac6c45616 / publicRT successfully

rtb-024d66a4ac6c45616 / publicRT

Details Info

Route table ID rtb-024d66a4ac6c45616	Main No	Explicit subnet associations -	Edge associations -
VPC vpc-01438603bad60b405 my-vpc-01	Owner ID 299582146630		

Routes Subnet associations Edge associations Route propagation Tags Related resources

Routes (2)

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-08947989aa30b850c	Active	No	Create Route
198.162.0.0/16	local	Active	No	Create Route Table

7. Associate Public Subnet

1. Click → Subnet associations
2. Select → PublicSubnet
3. Save

The screenshot shows the AWS Management Console interface. The top navigation bar includes the AWS logo, a search bar, and the user's profile. The left sidebar shows the VPC dashboard with options for Virtual private cloud, Subnets, Route tables, Internet gateways, and Egress-only internet. The main content area displays the 'Route tables (1/3)' page. A table lists route tables with columns for Name, Route table ID, Explicit subnet associations, Edge associations, and Main. The 'publicRT' route table is selected. An 'Actions' dropdown menu is open, showing options like 'View details', 'Set main route table', 'Edit subnet associations', 'Edit edge associations', 'Edit route propagation', 'Edit routes', 'Manage tags', and 'Delete route table'. Below this, the 'Edit subnet associations' page is shown, titled 'Edit subnet associations' with the subtitle 'Change which subnets are associated with this route table.' It features a table of 'Available subnets (1/2)' with columns for Name, Subnet ID, IPv4 CIDR, IPv6 CIDR, and Route table ID. The 'public-subnet-01' is selected. A 'Selected subnets' section shows the selected subnet. At the bottom, there are 'Cancel' and 'Save associations' buttons.

Route tables (1/3)

Name	Route table ID	Explicit subnet associations	Edge associations	Main
-	rtb-09855a9ec4d3e985f	-	-	Yes
-	rtb-05263edb131b81b95	-	-	Yes
publicRT	rtb-024d66a4ac6c45616	-	-	No

Edit subnet associations
Change which subnets are associated with this route table.

Available subnets (1/2)

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
public-subnet-01	subnet-015068d839a347954	198.162.1.0/24	-	Main (rtb-09855a9ec4d3e985f)
private-subnet-01	subnet-0cd0b641fc33c5b09	198.162.2.0/24	-	Main (rtb-09855a9ec4d3e985f)

Selected subnets

subnet-015068d839a347954 / public-subnet-01

Cancel Save associations

2) Enable VPC peering for cross-region.

STEP 1: Create VPC in Mumbai Region

1. Login to AWS Console
2. Select Region: ap-south-1 (Mumbai)
3. Go to VPC Dashboard
4. Click Create VPC
5. Select VPC only
6. Name: Mumbai-VPC
7. CIDR: 10.0.0.0/16

Click Create

VPC → Create VPC

aws [Search] [Alt+S]

VPC > Your VPCs > Create VPC

Create VPC [Info](#)

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create [Info](#)
Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.

mumbai-vpc

IPv4 CIDR block [Info](#)
☒ IPv4 CIDR manual input
☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR
10.0.0.0/16
CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)
☒ No IPv6 CIDR block
☐ IPAM-allocated IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block
☐ IPv6 CIDR owned by me

search [Alt+S]

vpc-0b7408d4602f12dc3

✓ You successfully created vpc-0b7408d4602f12dc3 / mumbai-vpc

vpc-0b7408d4602f12dc3 / mumbai-vpc [Actions](#)

Details [Info](#)

VPC ID vpc-0b7408d4602f12dc3	State Available	Block Public Access Off	DNS hostnames Disabled
DNS resolution Enabled	Tenancy default	DHCP option set dopt-06f9ef1cc1e355905	Main route table rtb-0c220e03b6f8e2079
Main network ACL acl-0a08842f5a67cdb4a	Default VPC No	IPv4 CIDR 10.0.0.0/16	IPv6 pool -
IPv6 CIDR (Network border group) -	Network Address Usage metrics Disabled	Route 53 Resolver DNS Firewall rule groups -	Owner ID 299582146630
Encryption control ID -	Encryption control mode -		

[Resource map](#) | CIDRs | Flow logs | Tags | Related resources | Integrations

Resource map [Info](#) Show all details

STEP 2: Create VPC in Virginia Region

1. Change Region to us-east-1 (Virginia)
2. Go to VPC Dashboard
3. Click Create VPC
4. Name: Virginia-VPC
5. CIDR: 192.168.0.0/16
6. Click Create

You successfully created vpc-0a22ea1ac1492252f / virginia-vpc

vpc-0a22ea1ac1492252f / virginia-vpc

Details Info

VPC ID vpc-0a22ea1ac1492252f	State Available	Block Public Access Off	DNS hostnames Disabled
DNS resolution Enabled	Tenancy default	DHCP option set dopt-0cf74ab7503438a83	Main route table rtb-0c0aa2aa1905fc5f8
Main network ACL acl-0b43053d20016dd91	Default VPC No	IPv4 CIDR 192.168.0.0/16	IPv6 pool -
IPv6 CIDR (Network border group) -	Network Address Usage metrics Disabled	Route 53 Resolver DNS Firewall rule groups -	Owner ID 299582146630
Encryption control ID -	Encryption control mode -		

Resource map Info

Resource map | CIDRs | Flow logs | Tags | Related resources | Integrations

Show all details

STEP 3: Create Peering Connection

1. Go to Mumbai Region
2. Open VPC Dashboard
3. Click Peering Connections
4. Click Create Peering Connection
5. Name: Cross-Region-Peering
6. Select Requester VPC: Mumbai-VPC
7. Select Another Region
8. Choose us-east-1
9. Select Virginia-VPC ID
10. Click Create

A VPC peering connection pcx-02a0455d9aa239d98 / Cross-Region-Peering has been requested. Remember to change your region to us-east-1 to accept the peering connection.

pcx-02a0455d9aa239d98 / Cross-Region-Peering

Details Info

Requester owner ID 299582146630	Accepter owner ID 299582146630	VPC Peering connection ARN arn:aws:ec2:ap-south-1:299582146630:vpc-peering-connection/pcx-02a0455d9aa239d98
Peering connection ID pcx-02a0455d9aa239d98	Requester VPC vpc-0b7408d4602f12dc3 / mumbai-vpc	Accepter VPC vpc-0a22ea1ac1492252f
Status Initiating Request to 299582146630	Requester CIDRs 10.0.0.0/16	Accepter CIDRs -
Expiration time Friday, July 24, 2026 at 19:12:26 GMT+5:30	Requester Region Mumbai (ap-south-1)	Accepter Region N. Virginia (us-east-1)

DNS settings

Requester VPC (vpc-0b7408d4602f12dc3 / mumbai-vpc) Info

Allow accepter VPC to resolve DNS of hosts in requester VPC to private IP addresses
Disabled

STEP 4: Accept Peering Connection

1. Switch to Virginia Region
2. Open VPC Dashboard
3. Click Peering Connections
4. Select Cross-Region-Peering
5. Click Actions
6. Click Accept Request
7. Status becomes Active

The screenshot shows the AWS VPC console interface. At the top, the region is set to 'United States (N. Virginia)'. The 'Peering connections' page is displayed, showing a table with one entry: 'pcx-02a0455d9aa239d98' with a status of 'Pending acceptance'. An 'Actions' dropdown menu is open, showing options like 'View details', 'Accept request', 'Reject request', 'Edit DNS settings', 'Manage tags', and 'Delete peering connection'. The 'Accept request' option is highlighted.

Below the table, the details for the peering connection 'pcx-02a0455d9aa239d98' are shown. The status is 'Pending acceptance'. The details include:

- Requester owner ID: 299582146630
- Requester VPC: vpc-0b7408d4602f12dc3
- Requester CIDRs: 10.0.0.0/16
- Requester Region: us-east-1
- Accepter owner ID: 299582146630
- Accepter VPC: vpc-0a22ea1ac1492252f / virginia-vpc
- Accepter CIDRs: 192.168.0.0/16
- VPC Peering connection ARN: arn:aws:ec2:us-east-1:299582146630:vpc-peering-connection/pcx-02a0455d9aa239d98

A green notification banner at the bottom states: 'Your VPC peering connection (pcx-02a0455d9aa239d98) has been established. To send and receive traffic across this VPC peering connection, you must add a route to the peered VPC in one or more of your VPC route tables.' A button 'Modify my route tables now' is provided.

The details section is updated to show the status as 'Active'.

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ch

[Alt+S] Ask Amazon Q

United States (N. Virginia)

Srihari (299582146630)

Srihari

pcx-02a0455d9aa239d98

pcx-02a0455d9aa239d98

Actions

Details Info

Requester owner ID

299582146630

Peering connection ID

pcx-02a0455d9aa239d98

Status

Active

Expiration time

-

Acceptor owner ID

299582146630

Requester VPC

vpc-0b7408d4602f12dc3

Requester CIDRs

10.0.0.0/16

Requester Region

Mumbai (ap-south-1)

VPC Peering connection ARN

arn:aws:ec2:us-east-1:299582146630:vpc-peering-connection/pcx-02a0455d9aa239d98

Acceptor VPC

vpc-0a22ea1ac1492252f / virginia-vpc

Acceptor CIDRs

192.168.0.0/16

Acceptor Region

N. Virginia (us-east-1)

DNS

Route tables

Tags

DNS settings

Edit DNS settings

Requester VPC (vpc-0b7408d4602f12dc3) Info

aws

Search

[Alt+S] Ask Amazon Q

United States (N. Virginia)

Srihari (299582146630)

Srihari

VPC > Route tables > rtb-0c0aa2aa1905fc5f8 > Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin	
192.168.0.0/16	local	Active	No	CreateRouteTable	
168.150.0.0/20	Peering Connection	Active	No	CreateRoute	Remove
0.0.0.0/16	Peering Connection	Active	No	CreateRoute	Remove

Add route

Cancel

Preview

Save changes

```
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

c2-user@ip-10-0-1-70 ~$ ping 10.0.1.70
PING 10.0.1.70 (10.0.1.70) 56(84) bytes of data:
bytes from 10.0.1.70: icmp_seq=1 ttl=127 time=0.018 ms
bytes from 10.0.1.70: icmp_seq=2 ttl=127 time=0.032 ms
bytes from 10.0.1.70: icmp_seq=3 ttl=127 time=0.033 ms
bytes from 10.0.1.70: icmp_seq=4 ttl=127 time=0.032 ms
bytes from 10.0.1.70: icmp_seq=5 ttl=127 time=0.032 ms
bytes from 10.0.1.70: icmp_seq=6 ttl=127 time=0.033 ms
bytes from 10.0.1.70: icmp_seq=7 ttl=127 time=0.030 ms
bytes from 10.0.1.70: icmp_seq=8 ttl=127 time=0.031 ms
bytes from 10.0.1.70: icmp_seq=9 ttl=127 time=0.032 ms
bytes from 10.0.1.70: icmp_seq=10 ttl=127 time=0.031 ms
bytes from 10.0.1.70: icmp_seq=11 ttl=127 time=0.033 ms
bytes from 10.0.1.70: icmp_seq=12 ttl=127 time=0.030 ms
bytes from 10.0.1.70: icmp_seq=13 ttl=127 time=0.034 ms
bytes from 10.0.1.70: icmp_seq=14 ttl=127 time=0.034 ms
bytes from 10.0.1.70: icmp_seq=15 ttl=127 time=0.031 ms
```

3) Enable VPC peering for cross-account (you can collaborate with your friend to do this task).

Cross-Account VPC Peering allows two VPCs from different AWS accounts to communicate privately.

EXAMPLE USED

Your Account ID: 111111111111

Friend Account ID: 222222222222

Your VPC:

Region: ap-south-1

CIDR: 10.0.0.0/16

VPC ID: vpc-xxxx

Friend VPC:

Region: us-east-1

CIDR: 192.168.0.0/16

VPC ID: vpc-yyyy

STEP 1: Get Friend Details

Ask friend to provide:

- Account ID
- VPC ID
- Region
- CIDR block

STEP 2: Create Peering from Your Account

1. Login to AWS Console
2. Go to VPC Dashboard
3. Click Peering Connections
4. Click Create Peering Connection
5. Name: Cross-Account-Peering
6. Requester VPC: Select your VPC
7. Account: Select Another account
8. Enter Friend Account ID
9. Select Friend Region
10. Enter Friend VPC ID (vpc-xxxx)
11. Click Create

[Alt+S] ⓘ ⓘ ⓘ ⓘ Asia Pacific (Mumbai) Srihari (299582146630) Srihari

> pcx-0ccaaf1d033494d9e ⓘ ⓘ

🟢 A VPC peering connection **pcx-0ccaaf1d033494d9e / abu-vpc-peering** has been requested. The owner of vpc-0e4318e00f5db6cae must accept the peering connection. ✕

pcx-0ccaaf1d033494d9e / abu-vpc-peering Actions ▾

Details Info

Requester owner ID 🔗 299582146630	Accepter owner ID 🔗 781436989524	VPC Peering connection ARN 🔗 arn:aws:ec2:ap-south-1:299582146630:vpc-peering-connection/pcx-0ccaaf1d033494d9e
Peering connection ID 🔗 pcx-0ccaaf1d033494d9e	Requester VPC 🔗 vpc-0b7408d4602f12dc3 / mumbai-vpc	Accepter VPC 🔗 vpc-0e4318e00f5db6cae
Status 🔄 Initiating Request to 781436989524	Requester CIDRs 🔗 10.0.0.0/16	Accepter CIDRs -
Expiration time Saturday, July 25, 2026 at 18:30:24 GMT+5:30	Requester Region Mumbai (ap-south-1)	Accepter Region Ohio (us-east-2)

DNS Route tables Tags

DNS settings Edit DNS settings

Requester VPC ([vpc-0b7408d4602f12dc3 / mumbai-vpc](#)) Info

STEP 3: Friend Accept Peering

Friend must:

1. Login to AWS Console
2. Go to VPC Dashboard
3. Click Peering Connections
4. Select Cross-Account-Peering
5. Click Actions
6. Click Accept Request
7. Status becomes Active

Accept VPC peering connection request Info ✕

Are you sure you want to accept this VPC peering connection request? (pcx-0ccaaf1d033494d9e)

Requester VPC 🔗 vpc-0b7408d4602f12dc3	Accepter VPC 🔗 vpc-0e4318e00f5db6cae / my-vpctask-ohio	Requester CIDRs 🔗 10.0.0.0/16
Accepter CIDRs -	Requester Region Mumbai (ap-south-1)	Accepter Region Ohio (us-east-2)
Requester owner ID 🔗 299582146630	Accepter owner ID 🔗 781436989524(This account)	

Cancel Accept request

[Alt+S] Ask Amazon Q United States (Ohio) user@26 (781436989524) user%4026

✔ Your VPC peering connection (pcx-0ccaaf1d033494d9e) has been established. To send and receive traffic across this VPC peering connection, you must add a route to the peered VPC in one or more of your VPC route tables. [Modify my route tables now](#)

Peering connections (1/3) [Info](#) [Actions](#) [Create peering connection](#)

Find peering connections by attribute or tag

Name	Peering connection ID	Status	Requester VPC	Accepter VPC
-	pcx-0ccaaf1d033494d9e	Provisioning	vpc-0b7408d4602f12dc3	vpc-0e4318e00f5db6cae / my-...

STEP 4: Update Route Table in Your Account

1. Go to VPC Dashboard
2. Click Route Tables
3. Select Route Table
4. Click Edit Routes
5. Click Add Route
6. Destination: Friend CIDR (192.168.0.0/16)
7. Target: Peering Connection
8. Click Save

RWS [Alt+S] Search Asia Pacific (Mumbai) Srihari (299582146630) Srihari

VPC > Route tables > rtb-0c220e03b6f8e2079 > Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
169.172.0.0/24	Peering Connection	-	No	CreateRoute

[Add route](#) [Cancel](#) [Preview](#) [Save changes](#)

✔ Updated routes for rtb-0c220e03b6f8e2079 successfully [Details](#)

rtb-0c220e03b6f8e2079 [Actions](#)

Details [Info](#)

Route table ID rtb-0c220e03b6f8e2079	Main Yes	Explicit subnet associations -	Edge associations -
VPC vpc-0b7408d4602f12dc3 mumbai-vpc	Owner ID 299582146630		

[Routes](#) [Subnet associations](#) [Edge associations](#) [Route propagation](#) [Tags](#) [Related resources](#)

STEP 5: Update Route Table in Friend Account

1. Go to Route Tables
2. Click Edit Routes
3. Click Add Route
4. Destination: Your CIDR (10.0.0.0/16)
5. Target: Peering Connection
6. Click Save

Allow inbound traffic from other VPC CIDR

Launch EC2 in both VPCs and test ping using private IP.

Both AWS accounts VPCs can communicate privately successfully

```
SRIHARI@LAPTOP-N3TSS9I MINGW64 ~/Downloads
$ ssh -i "awshari.pem" ec2-user@13.203.197.117

#_
#### Amazon Linux 2023
#####|
###|
##|
V~'--> https://aws.amazon.com/linux/amazon-linux-2023
~/m/'

Last login: Sat Jul 18 14:15:37 2026 from 103.203.172.230
[ec2-user@ip-10-0-0-168 ~]$ ping 3.20.221.231
PING 3.20.221.231 (3.20.221.231) 56(84) bytes of data.
64 bytes from 3.20.221.231: icmp_seq=208 ttl=120 time=197 ms
64 bytes from 3.20.221.231: icmp_seq=209 ttl=120 time=197 ms
64 bytes from 3.20.221.231: icmp_seq=210 ttl=120 time=197 ms
64 bytes from 3.20.221.231: icmp_seq=211 ttl=120 time=200 ms
64 bytes from 3.20.221.231: icmp_seq=212 ttl=120 time=200 ms
64 bytes from 3.20.221.231: icmp_seq=213 ttl=120 time=197 ms
64 bytes from 3.20.221.231: icmp_seq=214 ttl=120 time=197 ms
64 bytes from 3.20.221.231: icmp_seq=215 ttl=120 time=197 ms
64 bytes from 3.20.221.231: icmp_seq=216 ttl=120 time=197 ms
64 bytes from 3.20.221.231: icmp_seq=217 ttl=120 time=197 ms
64 bytes from 3.20.221.231: icmp_seq=218 ttl=120 time=197 ms
64 bytes from 3.20.221.231: icmp_seq=219 ttl=120 time=197 ms
64 bytes from 3.20.221.231: icmp_seq=220 ttl=120 time=197 ms
64 bytes from 3.20.221.231: icmp_seq=221 ttl=120 time=197 ms
64 bytes from 3.20.221.231: icmp_seq=222 ttl=120 time=197 ms
64 bytes from 3.20.221.231: icmp_seq=223 ttl=120 time=197 ms
64 bytes from 3.20.221.231: icmp_seq=224 ttl=120 time=197 ms
64 bytes from 3.20.221.231: icmp_seq=225 ttl=120 time=197 ms
64 bytes from 3.20.221.231: icmp_seq=226 ttl=120 time=197 ms
64 bytes from 3.20.221.231: icmp_seq=227 ttl=120 time=197 ms
64 bytes from 3.20.221.231: icmp_seq=228 ttl=120 time=197 ms
64 bytes from 3.20.221.231: icmp_seq=229 ttl=120 time=197 ms
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64 bytes from 3.20.221.231: icmp_seq=237 ttl=120 time=197 ms
64 bytes from 3.20.221.231: icmp_seq=238 ttl=120 time=197 ms
64 bytes from 3.20.221.231: icmp_seq=239 ttl=120 time=197 ms
64 bytes from 3.20.221.231: icmp_seq=240 ttl=120 time=197 ms
64 bytes from 3.20.221.231: icmp_seq=241 ttl=120 time=197 ms
64 bytes from 3.20.221.231: icmp_seq=242 ttl=120 time=197 ms
64 bytes from 3.20.221.231: icmp_seq=243 ttl=120 time=197 ms
64 bytes from 3.20.221.231: icmp_seq=244 ttl=120 time=197 ms
```

4) Set up a VPC Transit Gateway.

Create:

- VPC-1
- VPC-2
- Transit Gateway
- Attach both VPCs to the Transit Gateway
- Update Route Tables
- Test connectivity

Step 1: Create Two VPCs

VPC-1

- Name: VPC-1
- CIDR: 10.0.0.0/16

Create one subnet:

- 10.0.1.0/24

Launch one EC2 instance.

VPC-2

- Name: VPC-2
- CIDR: 192.168.0.0/16

Create one subnet:

- 192.168.1.0/24

Launch one EC2 instance.

Step 2: Create a Transit Gateway

1. Open **AWS Console**
2. Go to **VPC**
3. In the left menu, click **Transit Gateways**
4. Click **Create Transit Gateway**

Fill in:

- Name tag: **My-TGW**
- Description: Optional
- Amazon ASN: Keep the default

- Auto accept shared attachments: Default
- Default Route Table Association: **Enable**
- Default Route Table Propagation: **Enable**

Click **Create Transit Gateway**.

Wait until the state changes to **Available**.

Transit gateways Info

Find transit gateway by attribute or tag

No transit gateways
You do not have any transit gateways in this region

Create transit gateway

You successfully created tgw-043267960bbe52dc9 / My-TGW.

Transit gateways (1/1) Info

Find transit gateway by attribute or tag

Name	Transit gateway ID	Owner ID	State
My-TGW	tgw-043267960bbe52dc9	299582146630	Available

Step 3: Attach VPC-1

1. In the left menu, click Transit Gateway Attachments
2. Click Create Transit Gateway Attachment

Select:

- Transit Gateway: My-TGW
- Attachment Type: VPC
- VPC: VPC-1
- Choose the subnet (e.g., 10.0.1.0/24)

Click Create Attachment.

Wait until the attachment becomes Available.

Step 4: Attach VPC-2

Repeat the same steps:

- Transit Gateway: My-TGW
- Attachment Type: VPC

- Select VPC-2
- Choose the subnet (e.g., 192.168.1.0/24)

Click Create Attachment.

Wait until it becomes Available.

Create transit gateway attachment [Info](#)

A transit gateway (TGW) is a network transit hub that interconnects attachments (VPCs and VPNs) within the same AWS account or across AWS accounts.

Details

Name tag - optional
Creates a tag with the key set to Name and the value set to the specified string.

transist-gateway-attachment

Transit gateway ID [Info](#)
tgw-043267960bbe52dc9

Attachment type [Info](#)
VPC

VPC attachment
Select and configure your VPC attachment.

☒ DNS support [Info](#)

☒ Security Group Referencing support [Info](#)

☐ IPv6 support [Info](#)

☐ Appliance Mode support [Info](#)

VPC ID
Select the VPC to attach to the transit gateway.

Step 5: Update VPC-1 Route Table

Go to:

VPC → Route Tables

Select the route table for VPC-1.

Click Edit Routes → Add Route

Destination:

192.168.0.0/16

Target:

Transit Gateway

Save changes.

Step 6: Update VPC-2 Route Table

Select the route table for VPC-2.

Add a route:

Destination:

10.0.0.0/16

Target:

Transit Gateway

Save changes.

aws

Search

[Alt+S]

📧

🔔

🔧

Asia Pacific (Mumbai)

Srihari (299582146630)

VPC

Route tables

rtb-0aa1cc3a3397b7dca

Edit routes

Edit routes

Destination

10.0.0.0/16

Target

local

Status

Active

Propagated

No

Route Origin

CreateRouteTable

192.168.0.0/16

X

Transit Gateway

tgw-043267960bbe52dc9

Add route

Remove

Cancel

Preview

Save changes

aws

Search

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🔔

🔧

Asia Pacific (Mumbai)

Srihari (299582146630)

VPC

Route tables

rtb-0889371f119409178

Edit routes

Edit routes

Destination

192.168.0.0/16

Target

local

Status

Active

Propagated

No

Route Origin

CreateRouteTable

10.0.0.0/16

X

Transit Gateway

tgw-043267960bbe52dc9

Add route

Remove

Cancel

Preview

Save changes

rch

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🔔

🔧

Asia Pacific (Mumbai)

Srihari (299582146630)

rtb-0889371f119409178

🔔

🔧

Updated routes for rtb-0889371f119409178 successfully

Details

X

rtb-0889371f119409178

Actions

Details

Info

Route table ID

rtb-0889371f119409178

VPC

vpc-016d1ddbfb932e5 | my-vpc-02

Main

Yes

Owner ID

299582146630

Explicit subnet associations

-

Edge associations

-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Related resources

Routes (2)

Both

Edit routes

Filter routes

Destination

Target

Status

Propagated

Route Origin

10.0.0.0/16

tgw-043267960bbe52dc9

Active

No

Create Route

192.168.0.0/16

local

Active

No

Create Route Table

Step 7: Configure Security Groups

Allow ICMP (Ping) between the VPCs.

Example inbound rule:

- Type: All ICMP - IPv4
- Source: 10.0.0.0/16 (for VPC-2)

For VPC-1, allow:

- Source: 192.168.0.0/16

arch [Alt+S] Asia Pacific (Mumbai) Srihari (299582146630) Srihari

sg-0355c442f6d1e2293

Inbound security group rules successfully modified on security group (sg-0355c442f6d1e2293 | launch-wizard-3)

Details

sg-0355c442f6d1e2293 - launch-wizard-3

Actions

Details

Security group name launch-wizard-3	Security group ID sg-0355c442f6d1e2293	Description launch-wizard-3 created 2026-07-18T 15:43:46.137Z	VPC ID vpc-016d1ddbbfab932e5
Owner 299582146630	Inbound rules count 2 Permission entries	Outbound rules count 1 Permission entry	

Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - new | Tags

Inbound rules (2) Manage tags Edit inbound rules

IP version	Type	Protocol	Port range	Source	Description
IPv4	All ICMP - IPv4	ICMP	All	10.0.0.0/16	-
IPv4	SSH	TCP	22	0.0.0.0/0	-

dh [Alt+S] Asia Pacific (Mumbai) Srihari (299582146630) Srihari

sg-04ebed5b1510d7459

Inbound security group rules successfully modified on security group (sg-0355c442f6d1e2293 | launch-wizard-3)

Details

sg-04ebed5b1510d7459 - launch-wizard-2

Actions

Details

Security group name launch-wizard-2	Security group ID sg-04ebed5b1510d7459	Description launch-wizard-2 created 2026-07-18T 15:26:23.287Z	VPC ID vpc-0df884d3e4f0efe00
Owner 299582146630	Inbound rules count 2 Permission entries	Outbound rules count 1 Permission entry	

Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - new | Tags

Inbound rules (2) Manage tags Edit inbound rules

IP version	Type	Protocol	Port range	Source	Description
IPv4	SSH	TCP	22	0.0.0.0/0	-
IPv4	All ICMP - IPv4	ICMP	All	192.168.0.0/16	-

Connect to one EC2 (using a Bastion Host, Internet Gateway, or SSM).

Step 1: Create an Internet Gateway

- Go to **VPC → Internet Gateways**
- Click **Create Internet Gateway**
- Give it a name (e.g., My-IGW)
- Click **Create**

Step 2: Attach it to VPC-1

- Select the Internet Gateway
- Click **Actions → Attach to VPC**
- Select **VPC-1**

Step 3: Update the Route Table

In the route table of the **public subnet**, add:

Destination	Target
0.0.0.0/0	Internet Gateway

The screenshot shows the AWS Management Console interface. The top navigation bar includes the AWS logo, a search bar, and the region 'Asia Pacific (Mumbai)'. The breadcrumb trail indicates the path: VPC > Internet gateways > Attach to VPC (igw-0dc354422c7e3d058). The main heading is 'Attach to VPC (igw-0dc354422c7e3d058)'. Below this, there's a section for 'VPC' with instructions to attach an internet gateway. A search bar for 'Available VPCs' shows 'vpc-0df884d3e4f0efe00' selected. At the bottom of this section are 'Cancel' and 'Attach internet gateway' buttons. Below this is a green success banner: 'Updated routes for rtb-0aa1cc3a3397b7dca successfully'. The main content area shows the 'rtb-0aa1cc3a3397b7dca' route table details. It includes fields for 'Route table ID', 'VPC', 'Main' (Yes), and 'Owner ID'. Below the details are tabs for 'Routes', 'Subnet associations', 'Edge associations', 'Route propagation', 'Tags', and 'Related resources'. The 'Routes' tab is active, showing a table with 3 routes. The table has columns for Destination, Target, Status, Propagated, and Route Origin. The first route is for destination 0.0.0.0/0, targeting igw-0dc354422c7e3d058, with status 'Active' and origin 'Create Route'. The second route is for 10.0.0.0/16, targeting 'local', with status 'Active' and origin 'Create Route Table'. The third route is for 192.168.0.0/16, targeting tgw-043267960bbe52dc9, with status 'Active' and origin 'Create Route'.

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0dc354422c7e3d058	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table
192.168.0.0/16	tgw-043267960bbe52dc9	Active	No	Create Route

1. Launch one EC2 in VPC-1.
2. Launch one EC2 in VPC-2.
3. Configure Security Groups to allow ICMP (ping).

➤ Now the request is **coming from VPC-1**.

So, in the **Security Group of EC2-1**, you allow:

- Type: All ICMP
- **Source:** 192.168.0.0/16

This means:

"Allow ping requests from any IP in VPC-2."

➤ Now the request is **coming from VPC-1**.

So, in the **Security Group of EC2-2**, you allow:

- Type: All ICMP
- **Source:** 10.0.0.0/16

This means:

"Allow ping requests from any IP in VPC-1."

Step 8: Test Connectivity

Connect to the EC2 instance in **VPC-1**.

Run:

ping <Private-IP-of-EC2-in-VPC2>

Successfully created VPC endpoint
vpce-0f0f4d198d31d71e4

Endpoints (1/1) Info

Find endpoints by attribute or tag

VPC endpoint ID : vpce-0f0f4d198d31d71e4 X Clear filters

Name	VPC endpoint ID	Endpoint type	Status	Service name
MY ENDPOINT	vpce-0f0f4d198d31d71e4	Gateway	Available	com.amazonaws.us-east-1.s3

vpce-0f0f4d198d31d71e4 / MY ENDPOINT

Details Route tables Policy Tags

Details

Endpoint ID vpce-0f0f4d198d31d71e4	Status Available	Creation time Wednesday, February 18, 2026 at 15:23:06 GMT+5:30	Endpoint type Gateway
VPC ID vpc-0bdf9a1814e89b15f (MyVPC)	Status message -	Service name com.amazonaws.us-east-1.s3	Private DNS names enabled No
DNS record IP type service-defined	IP address type ipv4	Service region us-east-1	Private DNS preference -

VPC dashboard

AWS Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Endpoints (1/1) Info

Find endpoints by attribute or tag

VPC endpoint ID : vpce-0f0f4d198d31d71e4 X Clear filters

Name	VPC endpoint ID	Endpoint type	Status	Service name
MY ENDPOINT	vpce-0f0f4d198d31d71e4	Gateway	Available	com.amazonaws.us-east-1.s3

vpce-0f0f4d198d31d71e4 / MY ENDPOINT

Details Route tables Policy Tags

Details

Endpoint ID vpce-0f0f4d198d31d71e4	Status Available	Creation time Wednesday, February 18, 2026 at 15:23:06 GMT+5:30	Endpoint type Gateway
VPC ID vpc-0bdf9a1814e89b15f (MyVPC)	Status message -	Service name com.amazonaws.us-east-1.s3	Private DNS names enabled No
DNS record IP type service-defined	IP address type ipv4	Service region us-east-1	Private DNS preference -

Result:

Private EC2 can access S3 without Internet.